Distributed Optimization with Anytime Coupling Constraints and Asymptotic Consensus

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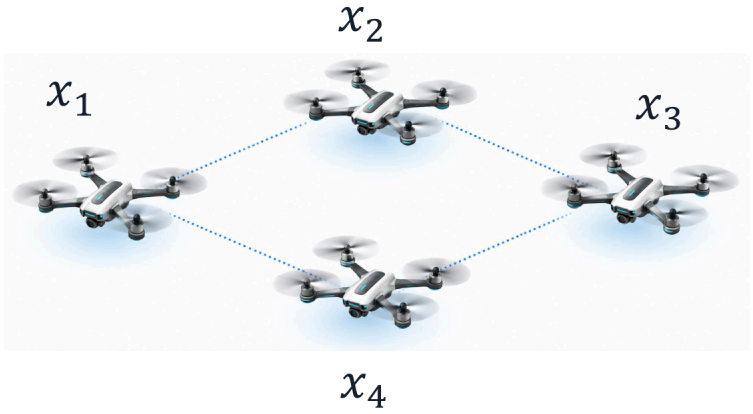
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Problem setting

Connected, undirected network



Local objective and constraints

Communication with neighbors.

$$x_i \in \mathbb{R}^n$$

Minimize total cost, strongly convex

$$F(x) = \sum_{i=1}^{N} f_i(x_i)$$

Coupling inequalities, convex

$$G_k(x) = \sum_{i=1}^{N} g_i^k(x_i) \leq 0, \qquad k = 1, \ldots, p$$

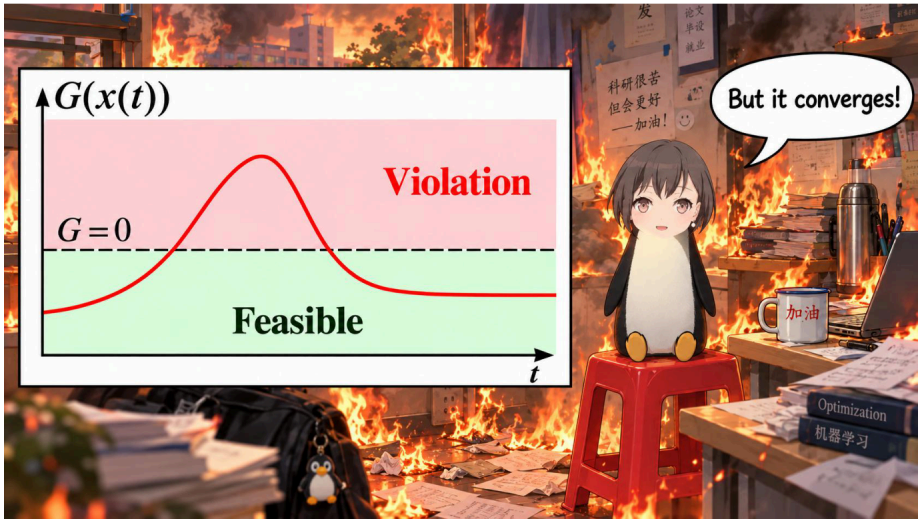
Consensus equality, affine

$$x_1 = \cdots = x_N \quad \Leftrightarrow \quad \mathcal{L}x = 0$$

$$x = \mathrm{col}(x_1, \ldots, x_N), \qquad \mathcal{L} = L \otimes I_n$$

Motivation

Operating limits and agreement have **different time requirements**.



The current output must remain usable during the whole process.

Anytime coupling feasibility

$$G_k(x(t)) \leq 0, \quad \forall t \geq 0$$

Asymptotic consensus

$$\mathcal{L}x(t) \to 0$$

Initialization: $G_k(x(0)) \leq 0$ while $\mathcal{L}x(0) \neq 0$ is allowed

Safe gradient flow

$$\min_{x \in \mathbb{R}^n} f(x) \qquad \text{s.t.} \quad g_k(x) \leq 0, \quad k = 1, \ldots, p$$

To solve the problem above, design

$$\dot{x} = S_\alpha(x)$$

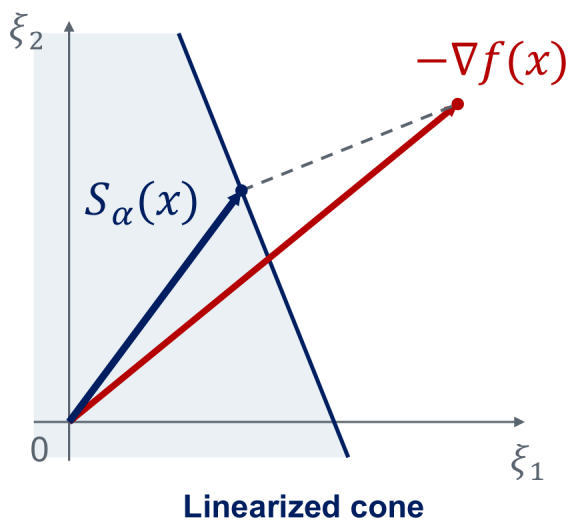
Safe gradient flow:

$$S_\alpha(x) = \arg\min_{\xi \in \mathbb{R}^n} \frac{1}{2} \|\xi + \nabla f(x)\|^2$$

subject to

$$\nabla g_k(x)^\top \xi \leq -\alpha g_k(x).$$

Velocity space $\dot{x}$



Linearized cone

Choose the admissible velocity closest to the desired descent.

A. Allibhoy and J. Cortés, "Control-Barrier-Function-Based Design of Gradient Flows for Constrained Nonlinear Programming," IEEE Transactions on Automatic Control, 2024.

Anytime feasibility

The safety margin determines the allowed rate of change.

Inside $g(x) < 0$



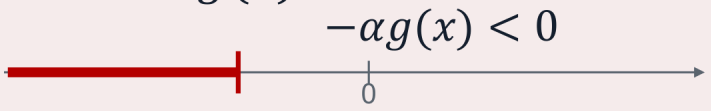
Use part of the available margin.

Boundary $g(x) = 0$



Block outward motion.

Outside $g(x) > 0$



Require the violation to decrease.

Comparison principle

$$\dot{g}(x) \leq -\alpha g(x)$$

$$\downarrow$$

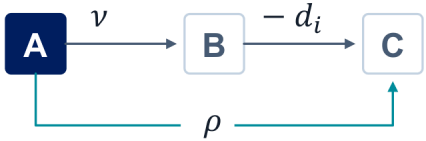
$$g(x(t)) \leq e^{-\alpha t} g(x(0))$$

A feasible start stays feasible.

Safe gradient flow: a well-defined, Lipschitz flow.

Distributed algorithm

1 cascade, 3 connected modules



A Auxiliary coordination (SP)

$$\tau\dot{\chi}_i = -\nabla f_i(\chi_i) - \sum_{k=1}^{p} \lambda_i^k \nabla g_i^k(\chi_i) - (\mathcal{L}\chi)_i - (\mathcal{L}\nu)_i$$

$$\tau\dot{\rho}_i^k = -\varepsilon\rho_i^k - (L\lambda^k)_i$$

$$\tau\dot{\lambda}_i^k = [g_i^k(\chi_i) + (L\rho^k)_i]_{\lambda_i^k}^{+}$$

$$\tau\dot{\nu}_i = (\mathcal{L}\chi)_i$$

ν

B Nominal direction

$$d_i = \nabla f_i(x_i) + (\mathcal{L}x)_i + (\mathcal{L}\nu)_i$$

$-d_i$

C Local SGF-QP

$$\dot{x}_i = \arg\min_{\xi_i \in \mathbb{R}^n} \frac{1}{2}\|\xi_i + d_i\|^2$$

subject to

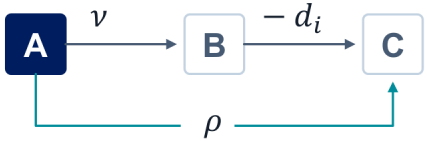
$$\nabla g_i^k(x_i)^\top \xi_i \le -\alpha\left[g_i^k(x_i) + (L\rho^k)_i\right], \quad k = 1, \ldots, p$$

ρ

$(\chi, \rho, \lambda, \nu)$ $\xrightarrow{\rho, \nu}$ **SGF-QP** $\xrightarrow{\dot{x}_i}$ $\int$ $\xrightarrow{x_i}$

Auxiliary coordination

A virtual optimizer, χ , learns the target and signals



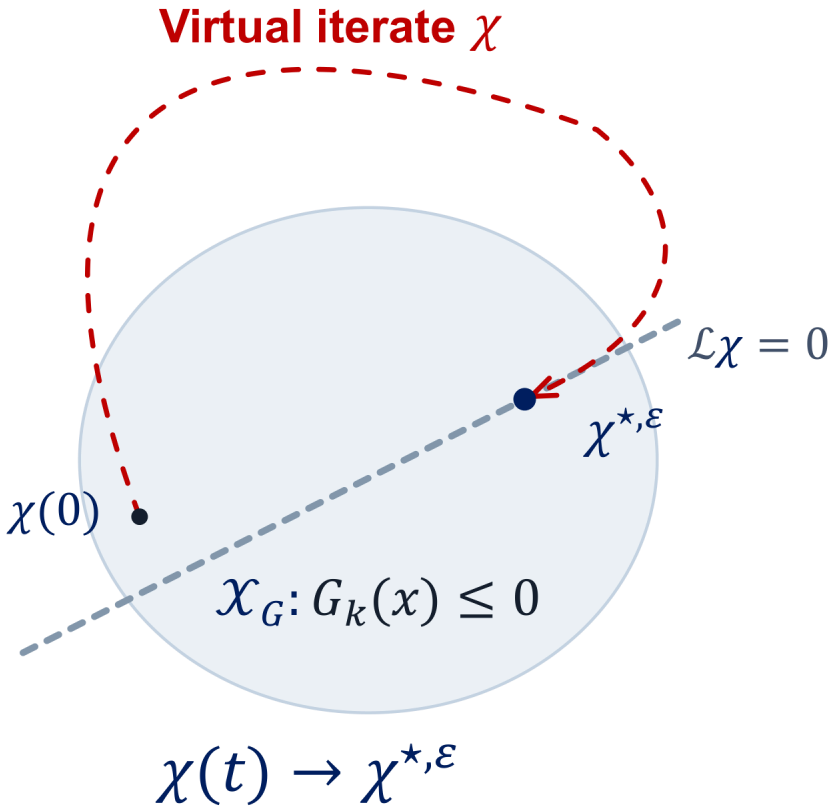
A Online coordination

$$\tau\dot{\rho}_i^k = -\varepsilon\rho_i^k - (L\lambda^k)_i$$

ρ ⟶ **local coupling limits in C**

$$\tau\dot{\nu}_i = (\mathcal{L}\chi)_i$$

ν ⟶ **nominal direction in B**



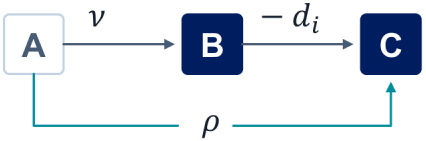
χ virtual optimization iterate

λ coupling-constraint multiplier

The virtual iterate may violate coupling limits during convergence.

From coordination to safe output

The direction and the safety limit meet in **1** QP program



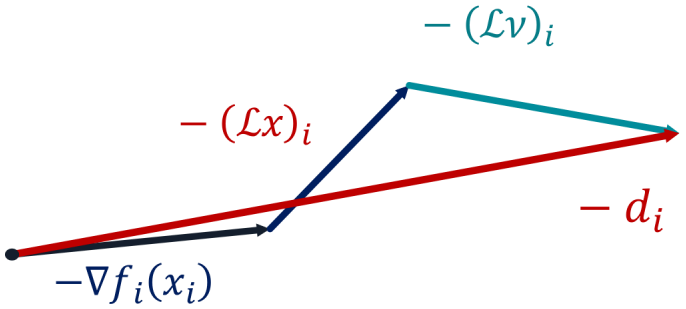
B Desired velocity

Mestres–Cortés: $-\nabla f_i(x_i)$

QP: inequalities and equalities

Ours:

$$-d_i = -\nabla f_i(x_i) - (\mathcal{L}x)_i - (\mathcal{L}\nu)_i$$



P. Mestres and J. Cortés, "Distributed and Anytime Algorithm for Network Optimization Problems with Separable Structure,"
IEEE Conference on Decision and Control, 2023.

C Safe output velocity

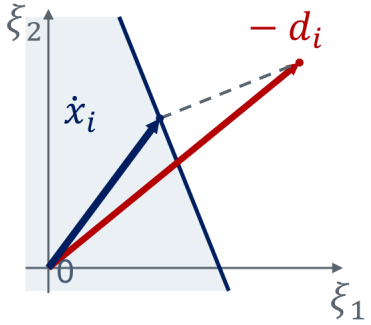
$$\dot{x}_i = \arg\min_{\xi_i \in \mathbb{R}^n} \frac{1}{2}\|\xi_i + d_i\|^2$$

subject to

$$\nabla g_i^k(x_i)^\top \xi_i \le -\alpha\left[g_i^k(x_i) + \left(L\rho^k\right)_i\right]$$

$$k = 1, \ldots, p$$

QP: localized coupling inequalities only



Global safety

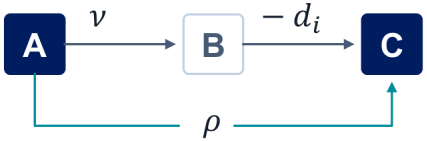
C

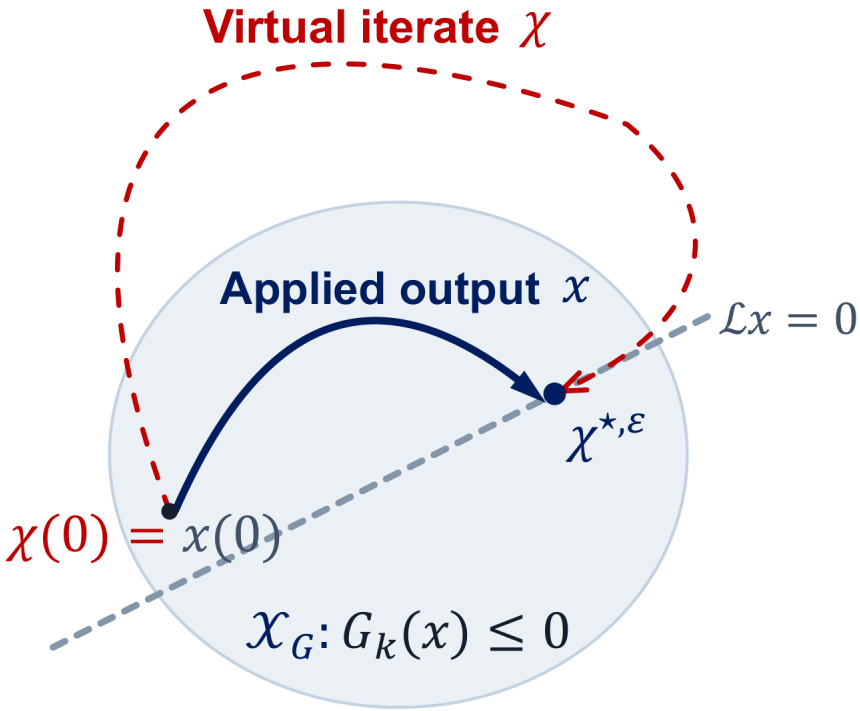
$$\dot{x}_i = \arg\min_{\xi_i \in \mathbb{R}^n} \frac{1}{2}\|\xi_i + d_i\|^2$$

subject to

$$\nabla g_i^k(x_i)^\top \dot{x}_i \leq -\alpha[g_i^k(x_i) + (L\rho^k)_i]$$

$$k = 1, \ldots, p$$





Sum over the network

$$\sum_{i=1}^{N} (L\rho^k)_i = \mathbf{1}_N^\top L\rho^k = 0$$

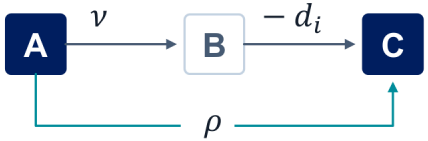
$$\therefore \dot{G}_k(x) \leq -\alpha G_k(x)$$

$$G_k(x(0)) \leq 0 \quad \Rightarrow \quad G_k(x(t)) \leq 0, \quad t \geq 0$$

Safety holds while the coordination signals are still evolving.

Parameters and convergence

Choose parameters at the modules where they act



A Auxiliary dynamics

$$\tau\dot{\rho}_i^k = -\varepsilon\rho_i^k - (L\lambda^k)_i$$

$\varepsilon \downarrow$ smaller target bias

$\tau \downarrow$ faster auxiliary coordination

C Output safety

$$\nabla g_i^k(x_i)^\top \xi_i \le -\alpha\left[g_i^k(x_i) + (L\rho^k)_i\right]$$

$\alpha > 0$ fix the safety-rate parameter

Practical guarantee

Choose a radius δ, then make the auxiliary layer fast enough (choose τ sufficiently small).

$$\limsup_{t\to\infty} \|x(t) - \chi^{\star,\varepsilon}\| \le \delta$$

Exact — stronger conditions

$$x(t) \to \chi^{\star,\varepsilon} \qquad \mathcal{L}x(t) \to 0$$

A bounded coupling-feasible set and **uniform** QP regularity.
Harder to verify globally.

Also note: $\|\chi^{\star,\varepsilon} - x^\star\| = O(\sqrt{\varepsilon})$

Numerical setup

Consider 5 agents on a unit-weight path



Agent	$(p_{i,1}, p_{i,2})$	$x_i(0)$
1	$(0,2)$	$(2,0)$
2	$(0.5,3)$	$(1,2)$
3	$(0.5,-1)$	$(-1,1)$
4	$(1,2)$	$(-2,2)$
5	$(1,3)$	$(1,1)$

$$x_i \in \mathbb{R}^2$$

$$f_i(x_i) = \frac{1}{2}\|x_i\|^2 - x_{i,1}$$

$$g_i(x_i) = 1 + p_{i,1}x_{i,1}^2 - p_{i,2}x_{i,2}$$

$$\alpha = 1, \quad \varepsilon = 10^{-3}, \quad \tau = 0.1$$

$$G(x(0)) = -1 < 0$$

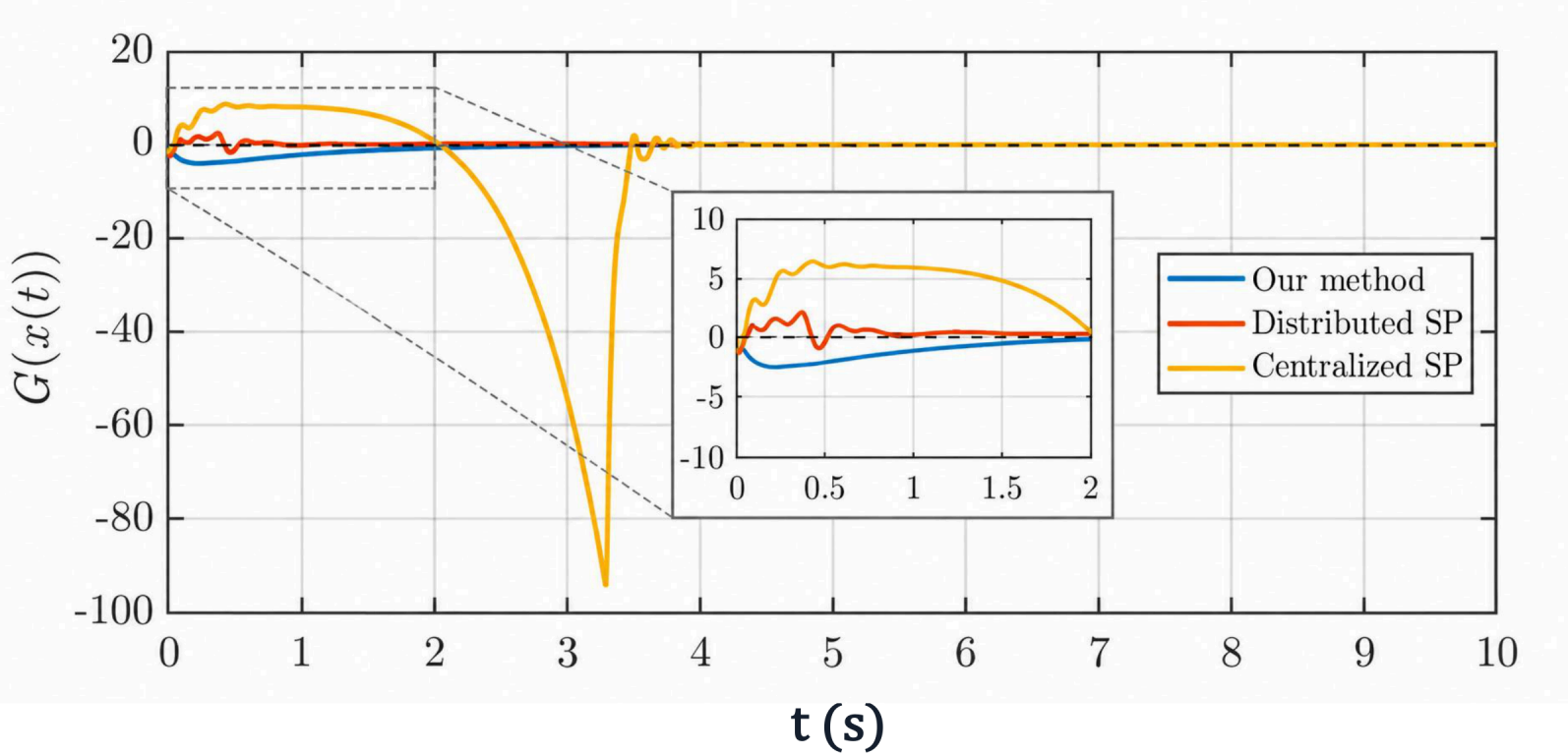
$$\|\mathcal{L}x(0)\| = \sqrt{50} > 0$$

$$\chi(0) = x(0), \qquad \rho(0) = \lambda(0) = \nu(0) = 0$$

Baselines: distributed SP and centralized SP (without SGF).

Coupling constraints

The baselines violate the shared limit during the transient.

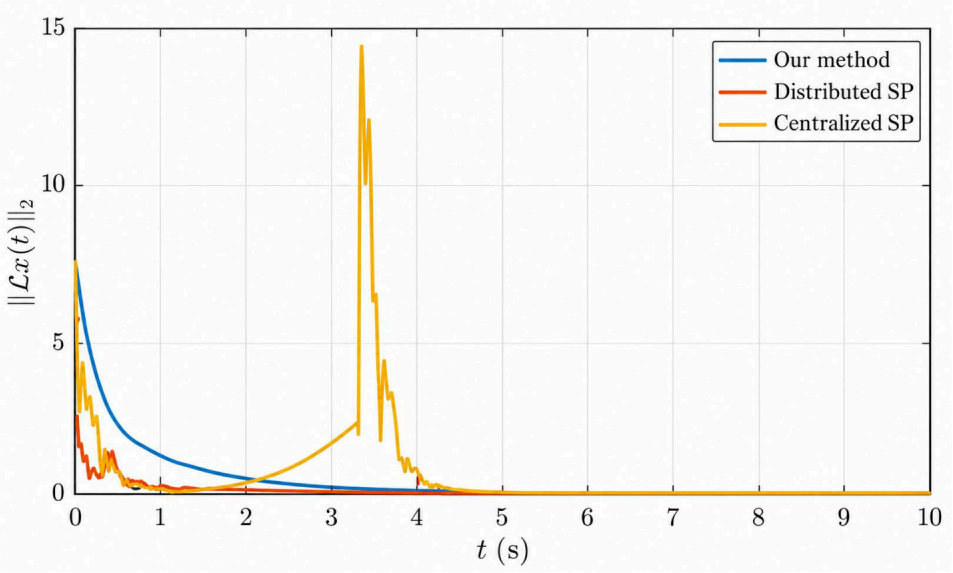


The proposed output stays on the feasible side of zero.

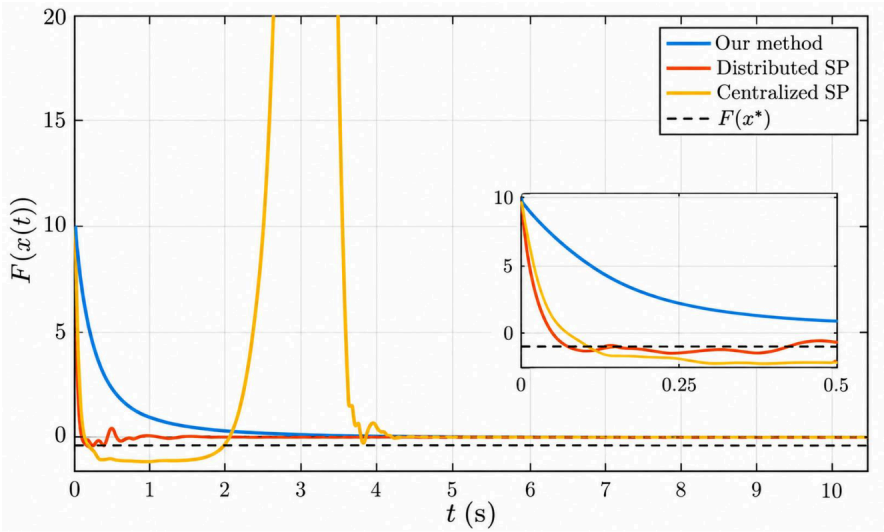
Consensus and objective

The three methods approach consensus in this numerical experiment.

Consensus error $||\mathcal{L}x(t)||$



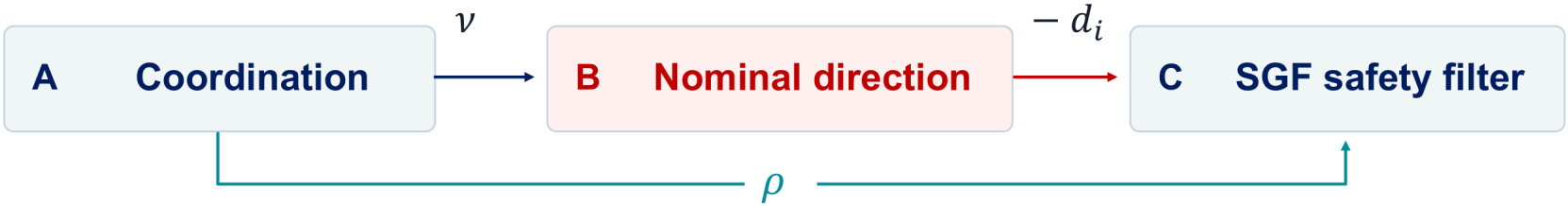
Objective value $F(x(t))$

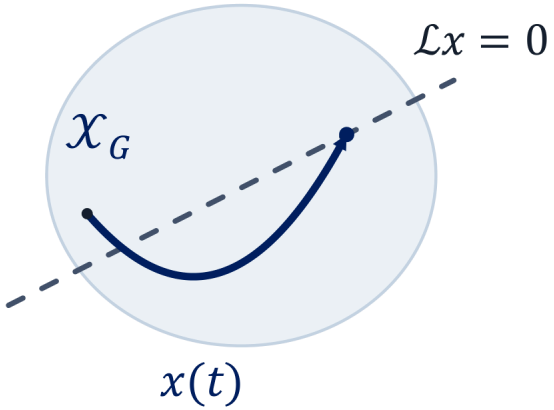


$F(x(t)) < F(x^\star)$ can occur before consensus.
However, this state is not yet fully feasible.

Conclusion

what we did? Coupling safety now, Consensus in the limit.





Anytime coupling feasibility

$$G_k(x(t)) \leq 0$$

Asymptotic consensus

$$\mathcal{L}x(t) \to 0$$

Under the stated convergence assumptions.

Outlook Hard anytime + asymptotic constraints ⟶ **General hybrid architectures**

Thank you!